

CHALLENGE 01

Task: Complete the installation and configuration of the following components:

1. AWS EC2 Instance with Jenkins
2. Apache Tomcat on Ubuntu VM
3. Docker Desktop on Windows

***AWS EC2 Instance with Jenkins**

- 1) Go to Aws console, Open EC2 and Launch Instance and
- 2) Give a Server Name ex: Jenkins-Server
- 3) Select Ubuntu OS Image
- 4) Select Number of Instance
- 5) Create new security group for server security
- 6) Open security group and

Allow ssh

Allow tcp

Allow 8080 port for Jenkins

And Launch the Instance Machine

The screenshot displays the AWS Management Console interface for an EC2 instance. The breadcrumb navigation at the top shows 'EC2 > Instances > i-030f9f9ff6dd1f7b1'. On the left, a sidebar contains navigation links for 'Dashboard', 'Global View', 'Instances', 'Instance Types', 'Launch Templates', 'Requests', 'Saved Plans', 'Saved Instances', 'Saved Hosts', 'Availability Reservations', 'Availability Manager', and 'Tags'. The main content area is titled 'Instance summary for i-030f9f9ff6dd1f7b1 (Jenkins)' and includes buttons for 'Connect', 'Instance state', and 'Actions'. Below the title, it states 'Updated less than a minute ago'. The summary is organized into several sections: 'Instance ID' (i-030f9f9ff6dd1f7b1), 'Public IPv4 address' (34.207.210.79), 'Private IPv4 addresses' (172.31.25.19), 'Instance state' (Running), 'Public DNS' (ec2-34-207-210-79.compute-1.amazonaws.com), 'Elastic IP addresses' (none), 'AWS Compute Optimizer finding' (Opt-in to AWS Compute Optimizer for recommendations), 'IPv6 address' (none), 'Instance type' (t3.micro), 'VPC ID' (vpc-05e67d56f859a1325), 'Hostname type' (IP name: ip-172-31-25-19.ec2.internal), 'Private IP DNS name (IPv4 only)' (ip-172-31-25-19.ec2.internal), 'Answer private resource DNS name' (IPv4 (A)), and 'Auto-assigned IP address' (34.207.210.79 [Public IP]).

- 7) Update package by using

\$ sudo apt update -y

- 8) Install java for Jenkins because java is the dependency of Jenkins

```

ubuntu@ip-172-31-25-19:~$ sudo apt update
sudo apt install fontconfig openjdk-21-jre
java -version
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
  fontconfig  openjdk-21-jre

Installing dependencies:
  adwaita-icon-theme  libatk-wrapper-java  libglx-mesa0  librsvg2-2  libxfixes3
  alsa-topology-conf  libatk-wrapper-java-jni  libglx0  librsvg2-common  libxft2
  alsa-ucm-conf  libatk1.0-0t64  libglibc-2-0  libsharpvuv0  libxi6

```

9) Install Jenkins from its official website by using

\$ wget <url>

\$ sudo apt update

```

ubuntu@ip-172-31-25-19:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 Packages [16.3 MB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages [248 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [64.2 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [31.2 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/main amd64 c-n-f Metadata [3984 B]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe amd64v3 Packages [126 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [39.2 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.2 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [3324 B]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64v3 Packages [219 kB]

```

\$ sudo apt install jenkins

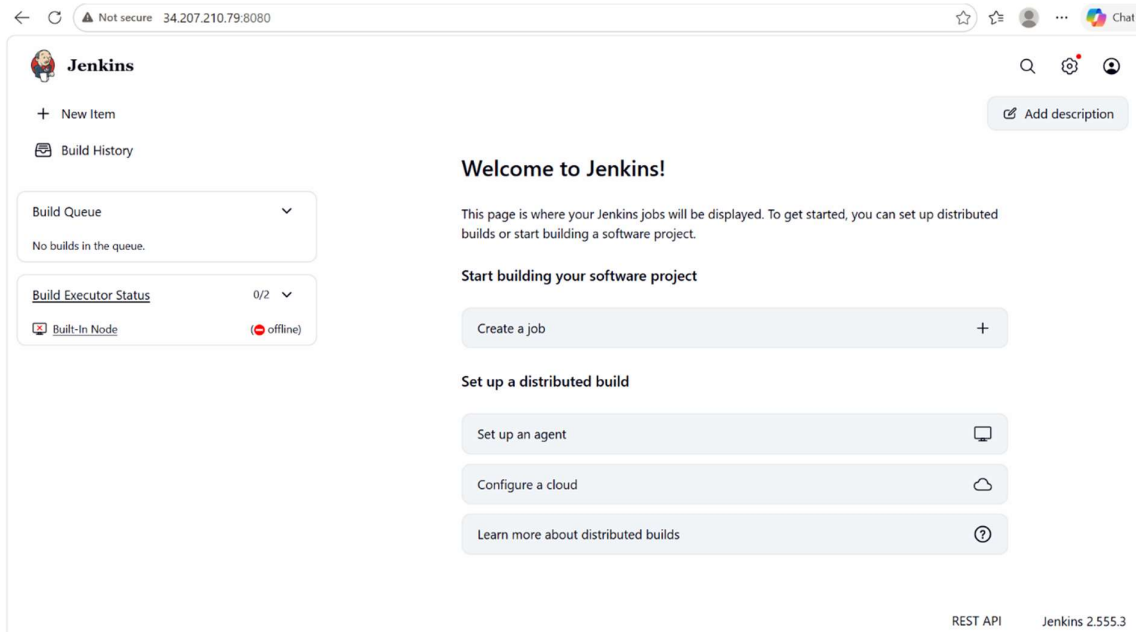
10) After installing start and enable the Jenkins by using the command

\$ sudo systemctl start Jenkins

\$ sudo systemctl enable Jenkins

\$ sudo systemctl status jenkins

11) After start and enable, copy the instance public ip address and past into the browser with
<ip>:8080



*Apache Tomcat on Ubuntu VM

1) Update the package

```
$ sudo apt update
```

2) Install java for depency

```
$ sudo apt install openjdk-17-jdk -y
```

```
vboxuser@Ubuntu:~/Desktop$ sudo apt install openjdk-17-jdk -y
Installing:
  openjdk-17-jdk

Installing dependencies:
  openjdk-17-jdk-headless  openjdk-17-jre  openjdk-17-jre-headless

Suggested packages:
  openjdk-17-demo  openjdk-17-source  visualvm  fonts-ipafont-gothic  fonts-ipafont-mincho  fonts-wqy-microhei |

Summary:
  Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 6
  Download size: 123 MB
  Space needed: 276 MB / 16.3 GB available

Get:1 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 openjdk-17-jre-headless amd64 17.0.19+10-1~26.04.2 [118 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 openjdk-17-jre amd64 17.0.19+10-1~26.04.2 [12.5 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 openjdk-17-jdk-headless amd64 17.0.19+10-1~26.04.2 [104 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 openjdk-17-jdk amd64 17.0.19+10-1~26.04.2 [12.5 kB]
Fetched 123 MB in 32s (3,860 kB/s)
Selecting previously unselected package openjdk-17-jre-headless:amd64.
```

3) Create a Parent directory for Tomcat

```
$ sudo mkdir -p /opt/tomcat
```

4) Go to browser and search Localhost:8080 in Ubuntu VM

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Apache Tomcat/10.1.56 

If you're seeing this, you've successfully installed Tomcat. Congratulations!



Recommended Reading:
[Security Considerations How-To](#)
[Manager Application How-To](#)
[Clustering/Session Replication How-To](#)

[Server Status](#)
[Manager App](#)
[Host Manager](#)

Developer Quick Start
[Tomcat Setup](#) [Realms & AAA](#) [Examples](#) [Servlet Specifications](#)
[First Web Application](#) [JDBC DataSources](#) [Tomcat Versions](#)

Managing Tomcat

For security, access to the [manager webapp](#) is restricted. Users are defined in:
\$CATALINA_HOME/conf/tomcat-users.xml

In Tomcat 10.1 access to the manager application is split between different users.

Documentation
[Tomcat 10.1 Documentation](#)
[Tomcat 10.1 Configuration](#)
[Tomcat Wiki](#)
Find additional important configuration

Getting Help
[FAQ and Mailing Lists](#)
The following mailing lists are available:
[tomcat-announce](#)
Important announcements, releases, security vulnerability notifications. (Low volume).

*Docker Desktop on Windows

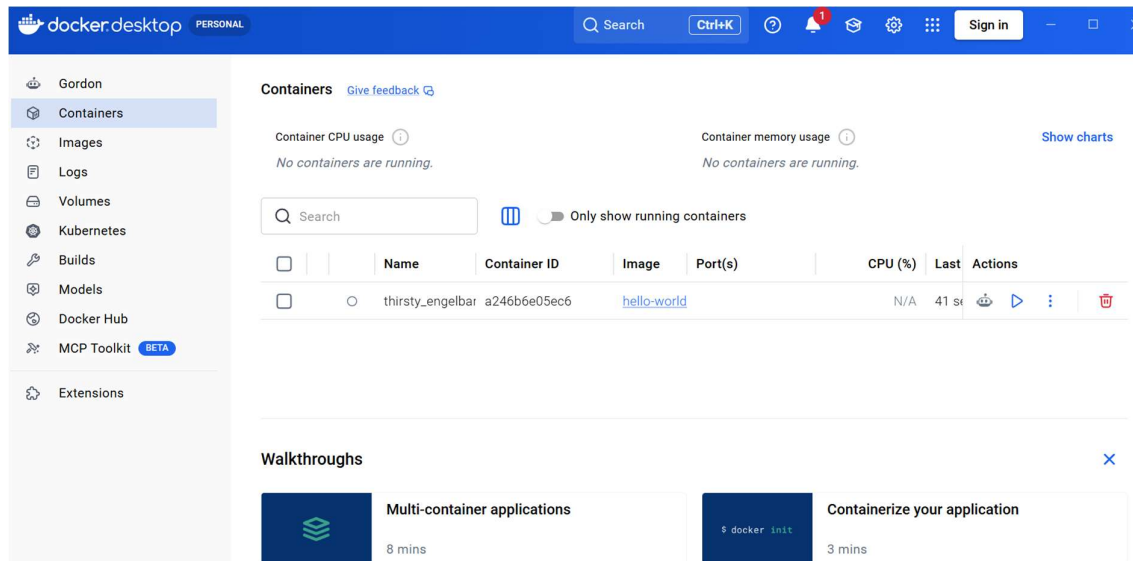
1) Open PowerShell as Administrator:

```
$ wsl --install --nodistribution
```

2) Download Docker Desktop From

<https://www.docker.com/products/docker-desktop/>

Install Docker Desktop using default settings.



3) Open PowerShell and Verify Docker Installation

\$ docker --version

4) Test

\$ docker run hello-world

Output: Hello from Docker

```
Administrator: Windows PowerShell

PS C:\WINDOWS\system32> docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
45e71e642bf5: Download complete
Digest: sha256:96498ffd522e70807ab6384a5c0485a79b9c7c08ca79ba08623edcad1054e62d
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```